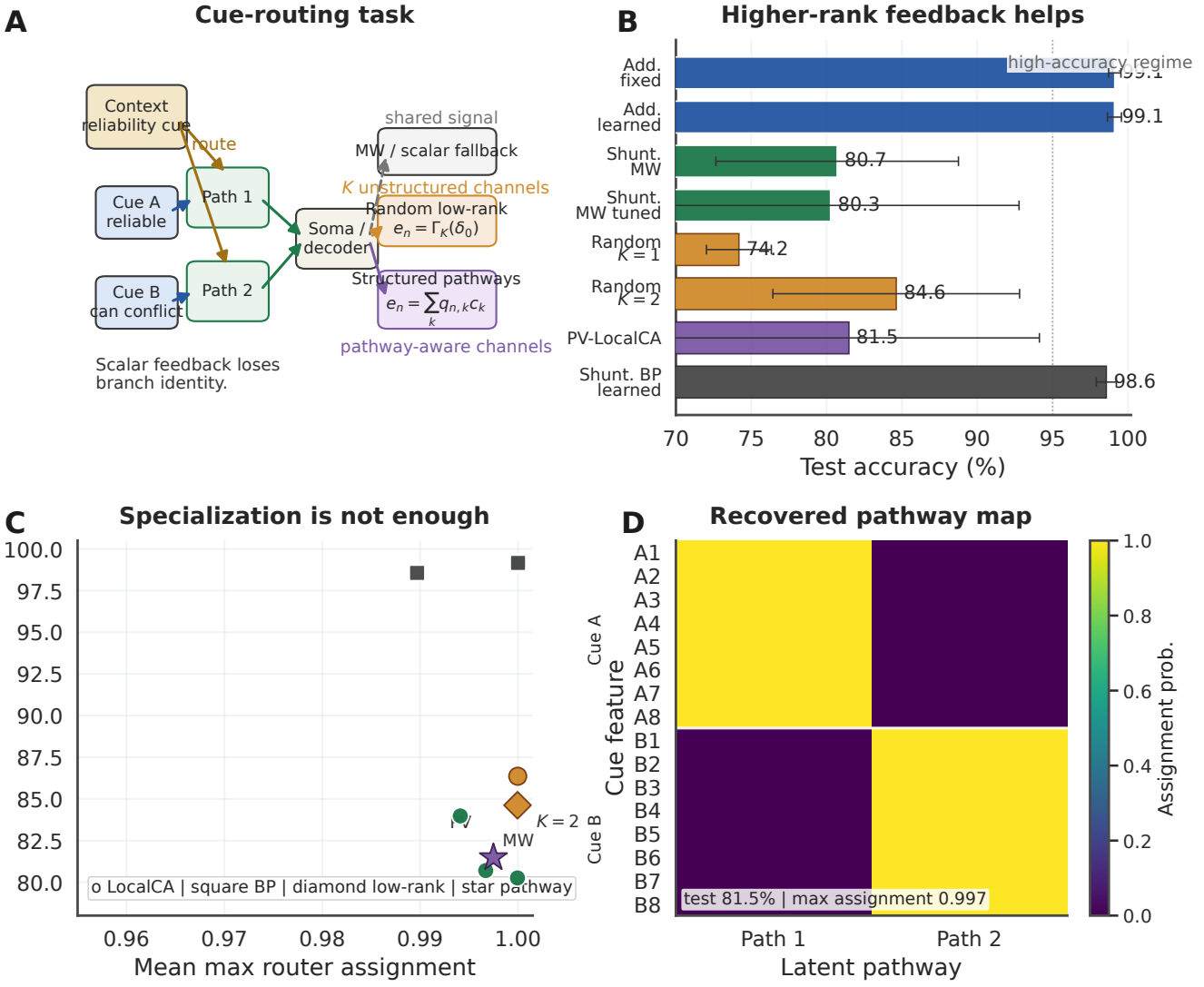

A
Cue-routing task
Context reliability cue
route
Cue A reliable
Path 1
Cue B can conflict
Path 2
Soma / decoder
shared signal
MW / scalar fallback
K unstructured channels
Random low-rank
$e_n = \Gamma_K(\delta_0)$
Structured pathways
$e_n = \sum_k q_{n,k} c_k$
pathway-aware channels
Scalar feedback loses branch identity.
B
Higher-rank feedback helps
high-accuracy regime
Add. fixed
Add. learned
99.1
Shunt. MW
80.7
Shunt. MW tuned
80.3
Random $K = 1$
74.2
Random $K = 2$
84.6
PV-LocalCA
81.5
Shunt. BP learned
98.6
70
75
80
85
90
95
100
Test accuracy (%)
C
Specialization is not enough
Test accuracy (%)
100.0
97.5
95.0
92.5
90.0
87.5
85.0
82.5
80.0
0.96
0.97
0.98
0.99
1.00
Mean max router assignment
$K = 2$
MW
o LocalCA | square BP | diamond low-rank | star pathway
D
Recovered pathway map
Cue feature
Cue A
Cue B
A1
A2
A3
A4
A5
A6
A7
A8
B1
B2
B3
B4
B5
B6
B7
B8
test 81.5% | max assignment 0.997
Path 1
Path 2
Latent pathway
Assignment prob.
1.0
0.8
0.6
0.4
0.2
0.0